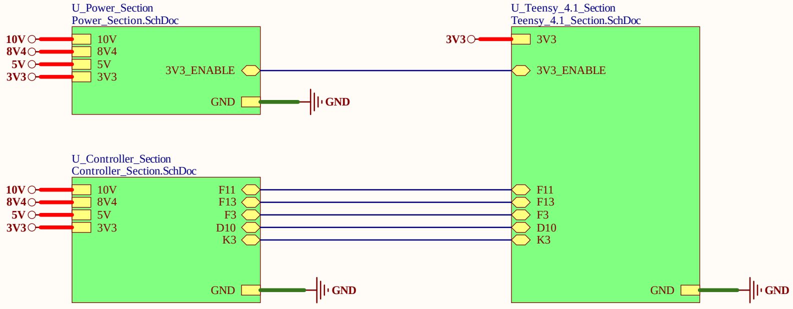
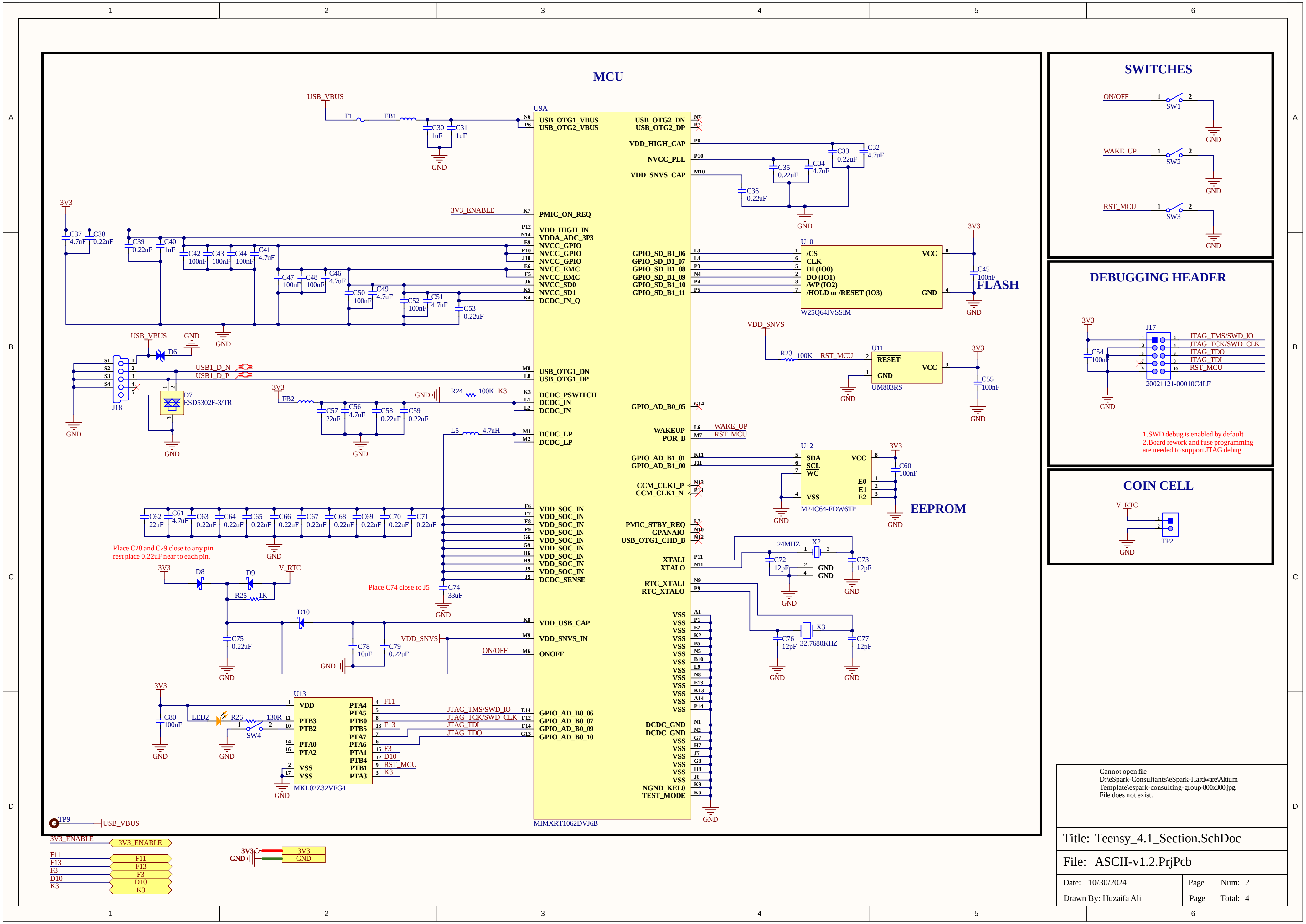

U_Power_Section
Power_Section.SchDoc
10V
8V4
5V
3V3
10V
8V4
5V
3V3
3V3_ENABLE
GND
GND
U_Teensy_4.1_Section
Teensy_4.1_Section.SchDoc
3V3
3V3
3V3_ENABLE
U_Controller_Section
Controller_Section.SchDoc
10V
8V4
5V
3V3
10V
8V4
5V
3V3
F11
F13
F3
D10
K3
GND
GND
F11
F13
F3
D10
K3
GND
GND

Cannot open file
D:\eSpark-Consultants\eSpark-Hardware\Altium
Template\espark-consulting-group-800x300.jpg.
File does not exist.

Title: Top_Section.SchDoc

File: ASCII-v1.2.PrjPcb

Date: 10/15/2024
Page Num: 1
Drawn By: Huzaifa Ali
Page Total: 4


MCU
U9A
USB_VBUS
F1
FB1
C30 1uF
C31 1uF
GND
USB_OTG1_VBUS
USB_OTG2_VBUS
USB_OTG2_DN
USB_OTG2_DP
VDD_HIGH_CAP
NVCC_PLL
VDD_SNVS_CAP
C33 0.22uF
C32 4.7uF
C35 0.22uF
C34 4.7uF
C36 0.22uF
3V3
3V3_ENABLE
PMIC_ON_REQ
VDD_HIGH_IN
VDDA_ADC_3P3
NVCC_GPIO
NVCC_GPIO
NVCC_GPIO
NVCC_EMC
NVCC_EMC
NVCC_SD0
NVCC_SD1
DCDC_IN_Q
C37 4.7uF
C38 0.22uF
C39 0.22uF
C40 1uF
C42 100nF
C43 100nF
C44 100nF
C41 4.7uF
C47 100nF
C48 100nF
C46 4.7uF
C50 100nF
C49 4.7uF
C52 100nF
C51 4.7uF
C53 0.22uF
GPIO_SD_B1_06
GPIO_SD_B1_07
GPIO_SD_B1_08
GPIO_SD_B1_09
GPIO_SD_B1_10
GPIO_SD_B1_11
U10
/CS
CLK
DI (IO0)
DO (IO1)
/WP (IO2)
/HOLD or /RESET (IO3)
VCC
GND
W25Q64JVSSIM
C45 100nF
FLASH
VDD_SNVS
R23 100K
RST_MCU
U11
RESET
GND
VCC
UM803RS
C55 100nF
USB_VBUS
GND
D6
S1
S2
S3
S4
J18
USB1_D_N
USB1_D_P
D7
ESD5302F-3/TR
USB_OTG1_DN
USB_OTG1_DP
FB2
C57 22uF
C56 4.7uF
C58 0.22uF
C59 0.22uF
R24 100K
K3
DCDC_PSWITCH
DCDC_IN
DCDC_IN
GPIO_AD_B0_05
L5 4.7uH
DCDC_LP
DCDC_LP
WAKEUP
POR_B
WAKE_UP
RST_MCU
GPIO_AD_B1_01
GPIO_AD_B1_00
U12
SDA
SCL
WC
VCC
E0
E1
E2
VSS
M24C64-FDW6TP
C60 100nF
EEPROM
CCM_CLK1_P
CCM_CLK1_N
C62 22uF
C61 4.7uF
C63 0.22uF
C64 0.22uF
C65 0.22uF
C66 0.22uF
C67 0.22uF
C68 0.22uF
C69 0.22uF
C70 0.22uF
C71 0.22uF
VDD_SOC_IN
DCDC_SENSE
PMIC_STBY_REQ
GPANAIO
USB_OTG1_CHD_B
24MHZ
X2
C72 12pF
C73 12pF
XTALI
XTALO
C74 33uF
V_RTC
D8
D9
R25 1K
RTC_XTALI
RTC_XTALO
D10
X3 32.7680KHZ
C76 12pF
C77 12pF
C75 0.22uF
VDD_USB_CAP
VDD_SNVS_IN
ONOFF
C78 10uF
C79 0.22uF
VDD_SNVS
ON/OFF
VSS
U13
VDD
PTB3
PTB2
PTA0
PTA2
VSS
VSS
PTA4
PTA5
PTB0
PTB5
PTA7
PTA6
PTA1
PTB4
PTB1
PTA3
MKL02Z32VFG4
C80 100nF
LED2
R26 130R
SW4
F11
F13
F3
D10
K3
JTAG_TMS/SWD_IO
JTAG_TCK/SWD_CLK
JTAG_TDI
JTAG_TDO
GPIO_AD_B0_06
GPIO_AD_B0_07
GPIO_AD_B0_09
GPIO_AD_B0_10
DCDC_GND
DCDC_GND
NGND_KEL0
TEST_MODE
MIMXRT1062DVJ6B
TP9
3V3_ENABLE
SWITCHES
ON/OFF
SW1
WAKE_UP
SW2
RST_MCU
SW3
DEBUGGING HEADER
J17
C54 100nF
JTAG_TMS/SWD_IO
JTAG_TCK/SWD_CLK
JTAG_TDO
JTAG_TDI
RST_MCU
20021121-00010C4LF
COIN CELL
V_RTC
TP2
1
2
3
4
5
6

Place C28 and C29 close to any pin rest place 0.22uF near to each pin.

Place C74 close to J5

1.SWD debug is enabled by default
2.Board rework and fuse programming are needed to support JTAG debug

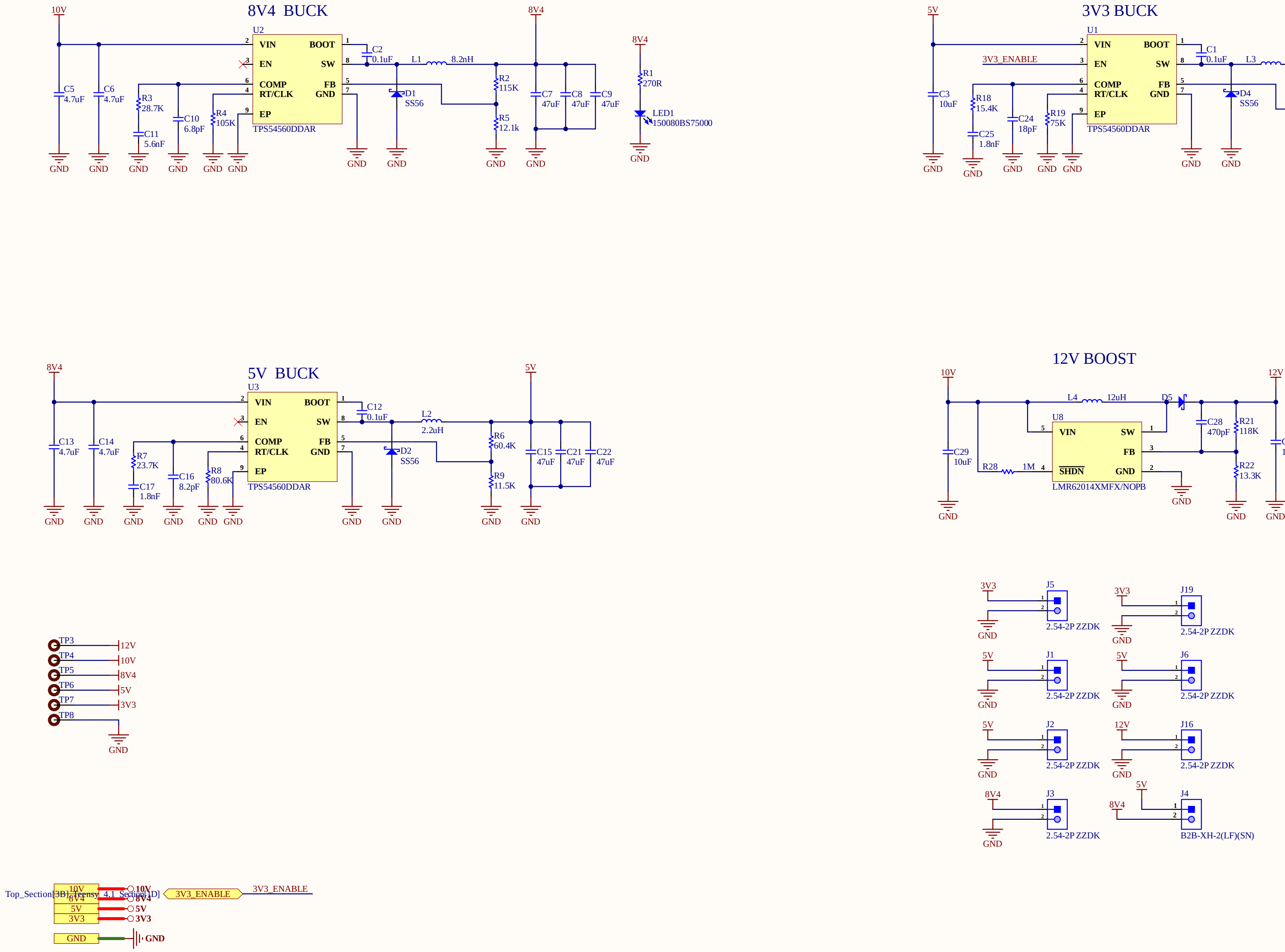
Cannot open file D:\eSpark-Consultants\eSpark-Hardware\Altium Template\espark-consulting-group-800x300.jpg. File does not exist.

Title: Teensy_4.1_Section.SchDoc

File: ASCII-v1.2.PrjPcb

Date: 10/30/2024 Page Num: 2

Drawn By: Huzaifa Ali Page Total: 4


8V4 BUCK
10V
U2
VIN
EN
COMP
RT/CLK
EP
BOOT
SW
FB
GND
TPS54560DDAR
C5 4.7uF
C6 4.7uF
R3 28.7K
C11 5.6nF
C10 6.8pF
R4 105K
C2 0.1uF
L1 8.2nH
D1 SS56
R2 115K
R5 12.1k
8V4
C7 47uF
C8 47uF
C9 47uF
R1 270R
LED1 150080BS75000
GND
3V3 BUCK
5V
U1
3V3_ENABLE
TPS54560DDAR
C3 10uF
R18 15.4K
C25 1.8nF
C24 18pF
R19 75K
C1 0.1uF
L3 1uH
D4 SS56
R17 36.5K
R20 11.5K
3V3
C4 47uF
C23 47uF
5V BUCK
8V4
U3
TPS54560DDAR
C13 4.7uF
C14 4.7uF
R7 23.7K
C17 1.8nF
C16 8.2pF
R8 80.6K
C12 0.1uF
L2 2.2uH
D2 SS56
R6 60.4K
R9 11.5K
5V
C15 47uF
C21 47uF
C22 47uF
12V BOOST
10V
L4 12uH
D5
U8
VIN
SW
FB
SHDN
GND
LMR62014XMFX/NOPB
C29 10uF
R28 1M
C28 470pF
R21 118K
R22 13.3K
12V
C27 10uF
TP3 12V
TP4 10V
TP5 8V4
TP6 5V
TP7 3V3
TP8 GND
3V3 J5 2.54-2P ZZDK
3V3 J19 2.54-2P ZZDK
5V J1 2.54-2P ZZDK
5V J6 2.54-2P ZZDK
5V J2 2.54-2P ZZDK
12V J16 2.54-2P ZZDK
8V4 J3 2.54-2P ZZDK
8V4 5V J4 B2B-XH-2(LF)(SN)
Top_Section[3B] Teensy_4.1_Section[1D]
10V
8V4
5V
3V3
GND
3V3_ENABLE

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Template\espark-consulting-group-800x300.jpg.
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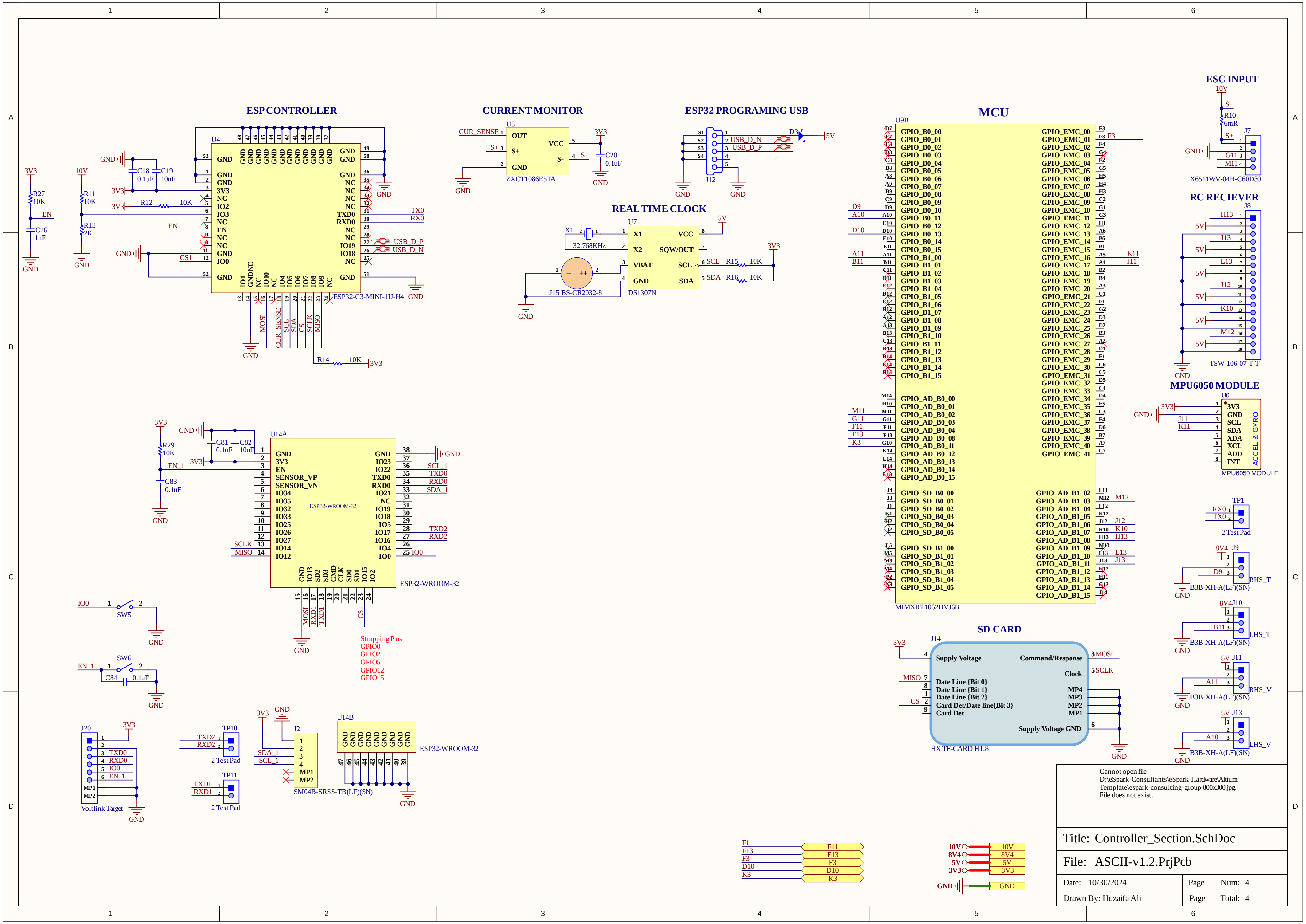
Title: Power_Section.SchDoc

File: ASCII-v1.2.PrjPcb

Date: 10/16/2024
Page Num: 3

Drawn By: Huzaifa Ali
Page Total: 4

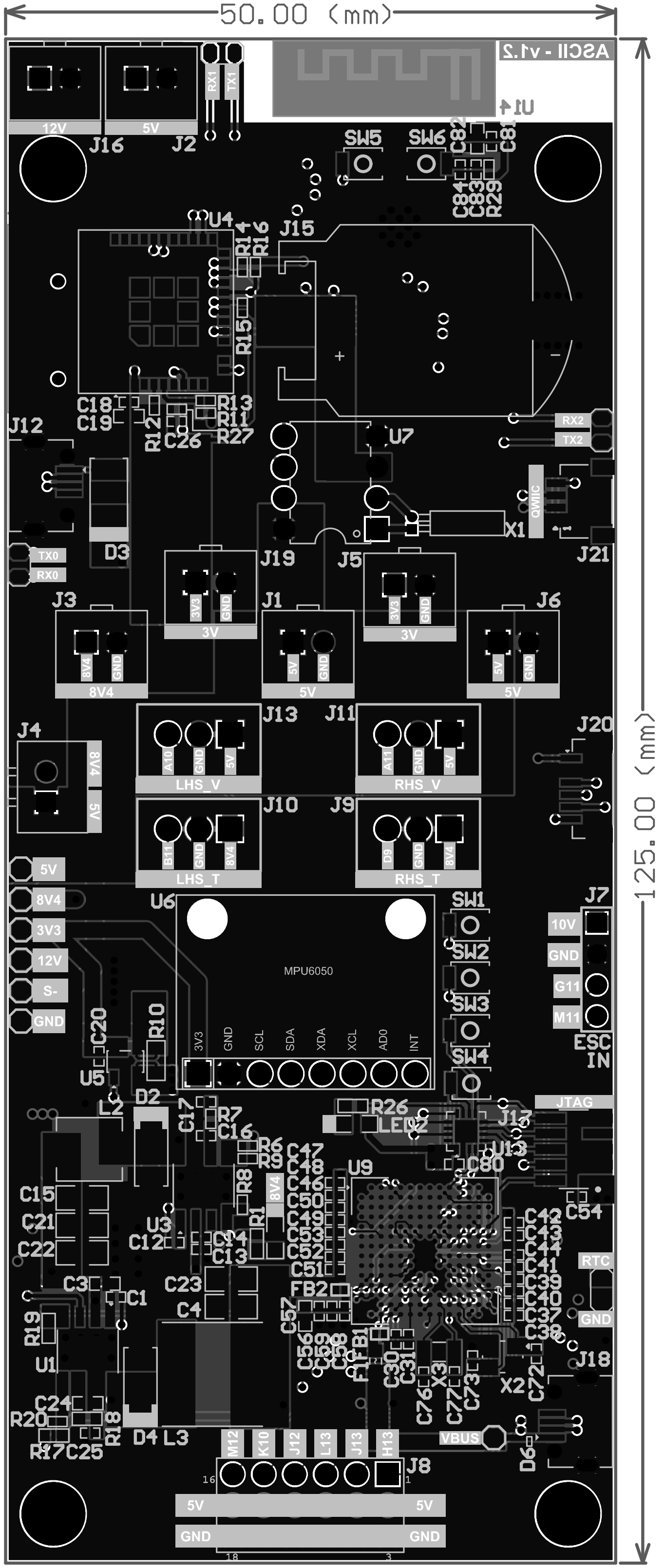

ESP CONTROLLER
U4
ESP32-C3-MINI-1U-H4
CURRENT MONITOR
U5
ZXCT1086E5TA
ESP32 PROGRAMING USB
J12
USB_D_N
USB_D_P
D3
5V
REAL TIME CLOCK
U7
DS1307N
X1
32.768KHz
J15 BS-CR2032-8
MCU
U9B
MIMXRT1062DVJ6B
ESC INPUT
10V
R10
6mR
J7
X6511WV-04H-C60D30
RC RECIEVER
J8
TSW-106-07-T-T
MPU6050 MODULE
U6
MPU6050 MODULE
ACCEL & GYRO
TP1
2 Test Pad
J9 RHS_T
J10 LHS_T
J11 RHS_V
J13 LHS_V
B3B-XH-A(LF)(SN)
U14A
ESP32-WROOM-32
U14B
Strapping Pins
GPIO0
GPIO2
GPIO5
GPIO12
GPIO15
SW5
SW6
J20
Voltlink Target
TP10
TP11
J21
SM04B-SRSS-TB(LF)(SN)
SD CARD
J14
HX TF-CARD H1.8
Supply Voltage
Command/Response
Clock
Date Line {Bit 0}
Date Line {Bit 1}
Date Line {Bit 2}
Card Det/Date line{Bit 3}
Card Det
Supply Voltage GND
GND
3V3
5V
8V4
10V

Cannot open file
D:\eSpark-Consultants\eSpark-Hardware\Altium Template\espark-consulting-group-800x300.jpg.
File does not exist.

Title: Controller_Section.SchDoc

File: ASCII-v1.2.PrjPcb

Date: 10/30/2024	Page Num: 4
Drawn By: Huzaifa Ali	Page Total: 4


50.00 (mm)
125.00 (mm)
ASCII - v1.2
U14
12V
5V
J16
J2
RX1
TX1
SW5
SW6
C82
C81
C84
C83
R29
U4
R14
R16
J15
R15
C18
C19
R13
R11
R27
R12
C26
J12
U7
RX2
TX2
QWIIC
X1
J21
D3
TX0
RX0
J19
J5
J3
J1
J6
3V3
GND
3V
8V4
5V
J4
J13
J11
J20
A10
A11
LHS_V
RHS_V
J10
J9
B11
D9
LHS_T
RHS_T
U6
MPU6050
3V3
GND
SCL
SDA
XDA
XCL
AD0
INT
SW1
SW2
SW3
SW4
J7
10V
GND
G11
M11
ESC IN
12V
S-
C20
R10
U5
L2
D2
C17
R7
C16
R26
LED2
J17
JTAG
U13
C80
R6
R9
C47
C48
C46
C50
C49
C53
C52
C51
U9
R8
R1
C15
C21
C22
U3
C12
C14
C13
C54
C42
C43
C44
C41
C39
C40
C37
C38
RTC
C3
C1
C23
C4
FB2
C57
C56
C55
C58
FB1
C30
C31
X3
C76
C77
C73
X2
C72
J18
R19
U1
C24
R20
R18
R17
C25
D4
L3
M12
K10
J12
L13
J13
H13
J8
VBUS
D6
16
1
18
3